

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2: Error Analysis Task

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| <b>COURSE NAME:</b> | Cloud Computing and<br>Big Data Analytics | <b>COURSE CODE:</b> | CSA1516 |
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### COURSE OUTCOME COVERED

**CO5:** *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*  
(BL5)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4)  
Question Count: 5

**Total Marks: 25**

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### Code of Conduct

I (**AMARA SATWIKA / 192525204**) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **A.SATWIKA**

### Assessment Tasks

#### Error Analysis Task

1. A Hadoop job repeatedly fails because input splits are uneven and reducers remain idle. Identify the likely design error and propose corrective actions.

#### **Uneven Input Splits and Idle Reducers.**

##### Answer

The main design error is improper input split and partitioning of data. When input splits have highly different sizes, some Mapper tasks take much longer than others. As a result, some reducers remain idle while waiting for mapper outputs.

##### Problems

- Uneven input file sizes.
- Poor input split configuration.
- Data skew between partitions.
- Improper number of reducers.
- Large mapper tasks delay the shuffle phase.

##### Corrective Actions

- Use an appropriate HDFS block size.
- Ensure input files are reasonably balanced.

- Use a suitable number of Mapper and Reducer tasks.
  - Apply a proper partitioner to distribute keys evenly.
  - Use a Combiner to reduce intermediate data.
  - Monitor task execution and identify data skew.
2. An HDFS deployment shows frequent data unavailability after a single node failure. Analyze the probable replication or NameNode design error.

### **HDFS Data Unavailability After Node Failure**

#### **Answer**

The probable error is insufficient data replication or a single-point-of-failure NameNode design.

HDFS normally stores multiple copies of each block. If only one copy exists on the failed node, the data becomes unavailable.

#### **Probable Causes**

- Replication factor set too low.
- Data replicas stored on the same node/rack.
- NameNode is not configured for High Availability.
- Improper replica placement.
- Failure detection or recovery is not properly configured.

#### **Corrective Actions**

- Set an appropriate **replication factor, preferably 3**.
- Distribute replicas across different DataNodes.
- Use **NameNode High Availability** with Active and Standby NameNodes.
- Enable automatic failover.
- Monitor DataNode health.
- Allow HDFS to automatically recreate missing replicas.

3. A data ingestion workflow using ETL and Apache NiFi creates duplicate records in the analytics store. Identify the likely causes and recommend fixes.

### **Duplicate Records in ETL and Apache NiFi**

#### **Answer**

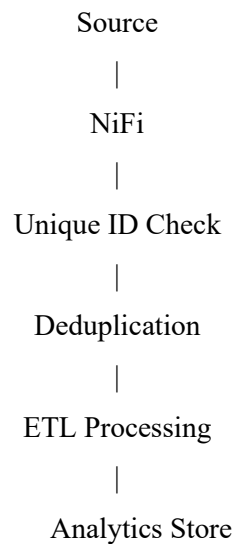
The main problem is duplicate data ingestion without proper record identification and processing control.

Probable Causes :

- Same source data is processed multiple times.
- NiFi flow is restarted without tracking processed records.
- No unique record ID.
- ETL process performs full extraction repeatedly.
- Database insertion does not check for existing records.
- Improper retry configuration.

**Corrective Actions :**

- Assign a unique ID to each record.
- Use incremental data extraction instead of full extraction.
- Maintain processing/checkpoint information.
- Use NiFi attributes to track processed data.
- Apply deduplication before storing data.
- Use database constraints such as unique keys.
- Configure retries carefully to avoid duplicate insertion.

**Correct flow :**

4. A YARN cluster suffers from resource starvation when multiple users submit jobs together. Analyze the scheduling error and suggest a better resource management approach.

**YARN Resource Starvation****Answer**

The likely scheduling error is improper resource allocation and lack of queue-based scheduling. One user's large jobs may consume most of the available cluster resources, leaving insufficient resources for other users.

**Problems**

- No resource limits.
- Poor scheduler configuration.
- One user consumes excessive CPU/memory.
- No separate queues.
- No priority or fairness policy.

**Corrective Actions :**

- Configure YARN Capacity Scheduler or Fair Scheduler.
- Create separate queues for different users or workloads.

- Set minimum and maximum resource limits.
- Assign appropriate priorities.
- Prevent a single job from consuming all resources.
- Monitor CPU and memory usage.

5. A backup strategy for distributed storage restores outdated data after recovery. Identify the probable design flaw in the replication or backup approach.

### **Outdated Data After Backup Recovery.**

#### **Answer**

The probable design flaw is using outdated or improperly synchronized backup copies. The backup system may not be capturing the latest changes before a failure occurs.

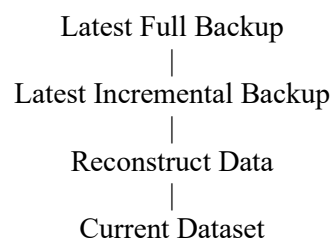
#### **Probable Causes**

- Backups are performed infrequently.
- Replicas are not synchronized.
- Backup contains old snapshots.
- No incremental backup mechanism.
- Recovery uses an older backup version.
- Backup metadata is not updated properly.

#### **Corrective Actions**

- Perform regular backups.
- Use incremental backups to capture recent changes.
- Maintain timestamped backup versions.
- Verify backup consistency.
- Synchronize replicas regularly.
- Use the latest valid snapshot during recovery.
- Test restoration periodically.

#### **Recovery Process :**



### Error Analysis Task Rubric

| Criteria                             | Weightage | Level 4 - Exemplary                                 | Level 3 - Proficient           | Level 2 - Developing       | Level 1 - Beginning     |
|--------------------------------------|-----------|-----------------------------------------------------|--------------------------------|----------------------------|-------------------------|
| Identification of Error              | 30%       | Accurately identifies the root technical issue      | Identifies most of the issue   | Partially identifies issue | Fails to identify issue |
| Cause Analysis                       | 20%       | Explains root cause with strong technical reasoning | Reasonable cause analysis      | Basic cause analysis       | Weak analysis           |
| Corrective Action                    | 25%       | Proposes effective and feasible corrections         | Reasonable corrections         | Basic corrections          | Ineffective corrections |
| Use of Hadoop / Integration Concepts | 15%       | Applies relevant concepts appropriately             | Mostly appropriate application | Partial application        | Incorrect application   |
| Clarity                              | 10%       | Clear and direct explanation                        | Generally clear                | Somewhat unclear           | Poorly presented        |